PermWorld: exact tasks, explicit relationships, auditable evidence

a A reusable mathematical task platform

Endpoint names, templates and instances are different counts

Platform v0.1: exact labels, shared interface

48 retained + 6 added = 54 task endpoints
48 canonical templates; 62 declared instances
56 materialized unary instances; one-line input
Versioned integer / structured-answer encoding

Three focus families

Family	Templates / instances
Cycle decomposition	6 / 8
Global subsequences	5 / 9
Permutation blocks	5 / 7

Focus: 16 templates / 24 instances / 22 nonconstant.
Counts do not establish mathematical independence.

b Relationships are explicit design objects

74 relation records; typed evidence and query conditions

Transformed-input relation

$\mathrm{LIS}(p) = \mathrm{LDS}(C(p))$
The source answer uses a complemented input.

Joint answers on the same input

$\mathrm{FP}(p) = n - \mathrm{exceedances}(p) - \mathrm{deficiencies}(p)$
Formula availability does not establish model use.

Exact generators → full-orbit isolated data

Paired training → adaptation → audited archive

Versioned complete-answer + EOS evaluation
Input isolation and legal task relations are separate checks.

Aliases and constants remain visible.
Missing relations mean unclassified, not unrelated.

c Data and compute have explicit audit trails

Archived platform audit · fixed inputs · paired training budgets

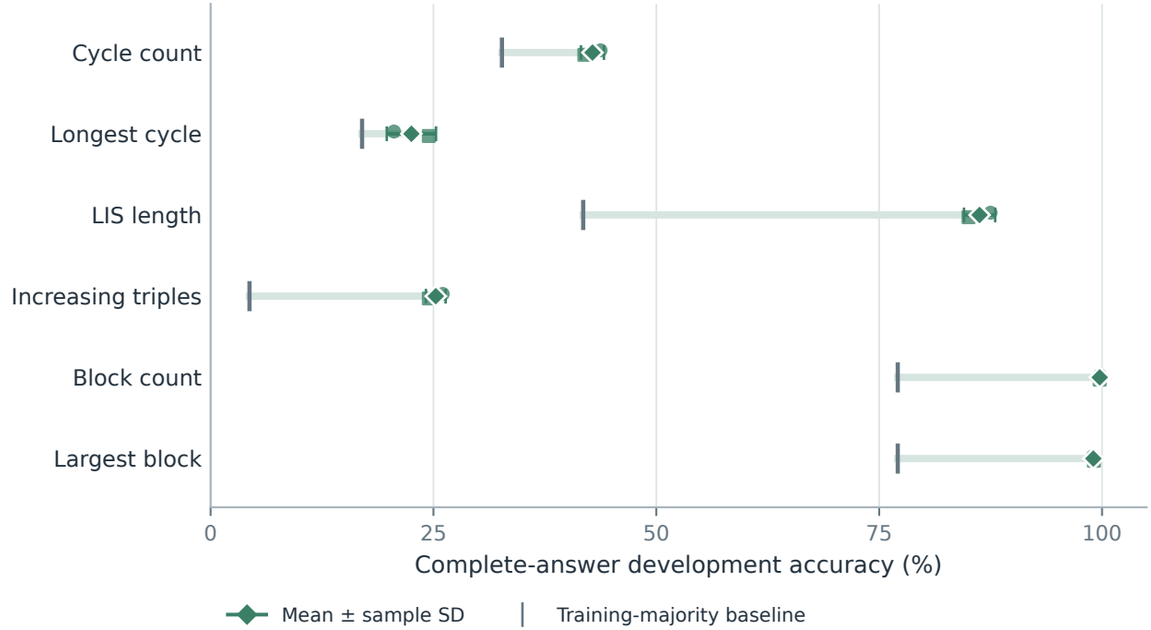
Input and label checks

9,216 inputs; 516,096 exact labels recomputed
0 duplicate inputs; 0 cross-role / repeat orbit overlaps
0 historical orbit overlaps; RNG streams replayed
Inverse, reversal and complement define isolation

Transfer protocol

Freeze tasks, input pools and selection rules
Account for sequence tokens and per-task exposure
Pair adaptation data, initialization and repeats
Evaluate full generated answers + EOS

Audit counts describe the archived unary dataset.
Binary-pair isolation was not certified; no new model run here.

d Learnability is measured, not assumed

Platform calibration · dev endpoint · 2 seeds · n=8,10,12